tions in order to establish the empirical relations between their various characteristics also helps us to answer a question that has been repeatedly raised in this book and in the literature in general; namely, whether hierarchical bureaucratic authority and professional expertness tend to go together, as Weber assumes, or whether they are alternative methods of administration and thus independent or perhaps even inversely related.[37]

A study by Stinchcombe compares the organization of construction and mass-production industries in the United States.[38] He reports that while construction industries are rationalized, just as are mass-production industries, they contain only a small bureaucratic apparatus through which the work process is hierarchically planned and governed, a situation which contrasts sharply with the organization of mass-production industries. Specifically, the proportion of clerical personnel in construction is far less than in other manufacturing industries; the proportion of authority positions occupied by professional administrators (administrators whose status derives from special education) is also much lower in construction; fewer specific directives flow from the superior (contractor) to the subordinate (the subcontractor);[39] and there is no career commitment to an organization, only to an occupation. This low degree of bureaucratization and rudimentary development of an administrative staff are probably due to the great seasonal variations of the operations in the construction industry. Evidence for this interpretation is furnished by comparisons within the construction industry: the higher the variation in seasonal employment in a given branch of the construction industry, the lower is the proportion of administrative staff (clerks) in the labor force of that branch. The formation of an administrative hierarchy and bureaucratic machinery requires continuity of personnel, but maintaining such a permanent staff is inefficient when an organization is faced by high seasonal fluctuations in volume and product mix. Stinchcombe argues that the greater degree of professionalization of its labor force

37 Reference has already been made to the work of Parsons and Gouldner in this connection. For other criticisms of Weber's concept, see Helen Constas, "Max Weber's Two Conceptions of Bureaucracy," *American Journal of Sociology,* 63 (1958), pp. 400–409; Roy G. Francis and Robert C. Stone, *Service and Procedure in Bureaucracy,* Minneapolis: University of Minnesota Press, 1956, pp. 153–159; and Eugene Litwak, "Models of Bureaucracy That Permit Conflict," *American Journal of Sociology,* 67 (1961), pp. 177–184.

38 Arthur L. Stinchcombe, "Bureaucratic and Craft Administration of Production," *Administrative Science Quarterly,* 4 (1959), pp. 168–187.

39 In construction, only the broad goals (the products and prices) are specified; in industrial organizations there is also specification of responsibilities and procedures (by whom and how the work is to be done).

enables the construction industry to function with a minimum of bureaucratization. The workers employed are for the most part skilled craftsmen, who can perform their tasks without much direction and control from superiors, because their work is guided by standards of craftsmanship which are akin to professional standards. In short, a professionalized labor force constitutes an alternative to the bureaucratic organization of work.

Stinchcombe concludes that his data suggest a refinement of Weber's classic conception of bureaucracy, which involves distinguishing rational from specifically bureaucratic administration. Rational administration entails primarily (1) a work force with specialized competence; (2) pecuniary compensation in return for contributions made, and (3) a contract that defines the goals or the broad responsibilities of the participants. Bureaucratic administration involves, in addition, (1) a hierarchy of authority through which the higher levels supervise the lower ones; (2) an administrative staff maintaining the lines of communication; and (3) continuity of operations and employment. Bureaucratic administration is one form of rational administration, but not the only one. The existence of a professionalized labor force—a group of skilled craftsmen enjoying continuity of occupational status in a labor market rather than in an organization—provides an alternative to bureaucratic organization as a means of rational administration.

Udy's study of 150 formally established production organizations in 150 different nonindustrial societies arrives independently at the same conclusion as Stinchcombe's comparison of two American industries.[40] Udy examines the relations between seven characteristics of these organizations by means of a correlation matrix. Three of the characteristics are designated as "bureaucratic": (1) hierarchical authority, referring to the existence of three or more levels of authority; (2) a specialized administrative staff whose members are not engaged directly in physical work; and (3) a set of rewards differentiated according to office. It is evident that the first two of these characteristics refer to the same concepts as Stinchcombe's, although the operational measures are, of course, not identical. The four characteristics labeled by Udy as "rational" are also similar to Stinchcombe's: (1) specialization, meaning that the organization is exclusively devoted to the production of goods; (2) rewards for performance related to contributions; (3) contractual agreements to define the terms of participation; and (4) dependence on superiors for rewards. While the fourth characteristic is not included in Stinchcombe's list describing rational administra-

[40] Stanley H. Udy, Jr., " 'Bureaucracy' and 'Rationality' in Weber's Organization Theory," *American Sociological Review,* 24 (1959), pp. 791–795.

tion, it is clearly a characteristic of the organizations he examined. Udy's analysis shows that the three bureaucratic characteristics were directly related to one another,[41] and five of the six direct relations between the four rational characteristics were also significant.[42] Moreover, none of the twelve correlations between one bureaucratic and one rational characteristic were significantly positive, and three of them (all involving the factor "specialized administrative staff") were significantly negative. In sum, the extent of bureaucratization and the degree of rationality in these organizations were largely independent (and, if anything, inversely related). Thus, two entirely different sets of data yield essentially the same results.

What then may we conclude about the relation between professionalization and bureaucratization? Professional work, as we have previously noted, is increasingly being carried out in large, complex organizations which are bureaucratically organized. This trend would seem to contradict Stinchcombe's conclusion that professionalization and bureaucratization are alternative methods of rational administration. The contradiction can be resolved by suggesting that not all aspects of bureaucratization are incompatible with the exercise of professional judgment, although one is, namely, hierarchical supervision and discipline. The other factors included in Stinchcombe's conception of bureaucracy—centralized management, an administrative staff, and continuity of employment—are probably not found in the construction industry because of the particular circumstances there; specifically, because they are incompatible with seasonal fluctuations in operations. But managerial planning and coordination, the administrative staff they necessitate, and continuity of employment are not inherently incompatible with decision-making on the basis of professional expertness; only disciplined compliance with orders of hierarchical superiors entails a fundamental conflict with professionalism. In the absence of direct hierarchical supervision, genuine professional work can be and, indeed, frequently is carried out in otherwise bureaucratized organizations. To say this, however, is not to deny that conflicts between professional and administrative considerations often arise in these organizations.

Technological Complexity. Durkheim's classic analysis of the division of labor in societies holds that increases in social density promote specialization, which, in turn, changes the basis of social solidarity from common values—"mechanical solidarity"—to interdependence with a greater tolerance for differing value orientations—"organic solidar-

[41] The Q values using Yule's coefficient of association were +.79, +.79, and +.70. *Ibid.*, p. 794.

[42] Q values were +.66, +1.00, +.75, +.72, and +.61. The relationship between specialization and dependence was positive but not significant (+.35). *Loc. cit.*

ity."[43] Some implications of Durkheim's analysis may be explored by considering the relation between hierarchical differentiation and specialization in primitive production organizations.

Udy's research on production organizations in nonindustrial societies reveals, as we just mentioned, that a minimum of specialization —the mere fact that an organization is exclusively devoted to production—was not associated with the existence of a bureaucratic hierarchy of, at least, three levels. But further specialization—the carrying out of three or more different operations concurrently—was positively related to such hierarchical differentiation in an organization.[44] Does this difference mean, as Udy implies, that specialization, by creating a need for coordination, leads to the development of a bureaucratic hierarchy? We submit that the direction of influence between the two variables may be the opposite and that some evidence from Udy's own research supports this alternative interpretation. He found bureaucratic hierarchies more prevalent in organizations where membership was based on kinship or political-status obligations than in those where it was based on voluntary agreement or contract.[45] This observation suggests that hierarchical differentiation in a simple production organization tends to be rooted in the status structure of the community rather than to emerge in response to the need for coordination in the organization and that, consequently, the correlation between hierarchical differentiation and specialization implies that the differentiation promotes specialization, not the reverse. This interpretation is also supported by Udy's finding, cited in the discussion of the evolution of organizational complexity, that specialization tends to emerge in organizations only when hierarchical dependence, and thus surely hierarchical differentiation itself, have already developed. A further inference one might draw is that power differences and the opportunity of some members to exploit others encourage the development of specialization in production, since such specialization often means that some participants have more desirable duties than others. A bit of evidence in support of this speculation is provided by Udy's finding that the existence of a multilevel hierarchy is associated with the fact that one or more superiors in the organization (members above the bottom level) do not work.[46]

[43] Emile Durkheim, *The Division of Labor in Society,* George Simpson (trans.), Glencoe, Ill.: Free Press, 1947, pp. 70–132, 256–282 (first published 1893).

[44] Stanley H. Udy, Jr., "The Structure of Authority in Non-Industrial Production Organizations," *American Journal of Sociology,* 64 (1959), pp. 582–584; and *The Organization of Work,* New Haven: Human Relations Area Files, 1959, p. 38.

[45] Stanley H. Udy, Jr., "Technology, Society and Production Organization," paper read at the meetings of the American Sociological Association, New York, 1960.

[46] Udy, *The Organization of Work, op. cit.,* p. 39.

These conclusions, although highly tentative, suggest some specification of intervening variables in Durkheim's theory. Perhaps increases in social density foster specialization because and if they lead to a greater differentiation of power in the community, which is reflected in the organization of production inasmuch as it enables some members to assign the most burdensome tasks to others.

We turn now from a consideration of rudimentary technological systems to an examination of organizations exhibiting advanced technology. There can be little doubt that increased mechanization in production organizations has resulted in higher productivity. Using installed electrical horsepower per production worker as an index of the degree of mechanization, and output per man-hour as an index of productivity, Melman shows that in the United States between 1899 and 1950 productivity increases paralleled increases in mechanization.[47] A comparison between countries at different levels of mechanization also reveals some relation between installed horsepower and productivity per worker.[48] But data from 15 British and American industries show no consistent relation between the measures of mechanization and productivity. Melman suggests that this finding does not negate the finding of a relation between mechanization and productivity but merely indicates the importance of other factors, such as type of organization, for industrial productivity.

Changes in mechanization, in turn, appear to be related to labor costs, specifically, to the cost of labor relative to that of machine production. For example, Melman shows that between 1938 and 1950 the cost of manual handling of materials in British firms increased more rapidly than the cost of handling them by machine (fork-lift and elevating-platform trucks). As labor costs increased, the volume of work at which machine operation became less expensive than manual operation declined. As a result, it became economical even for the firms with a relatively small volume of work to mechanize.[49] Melman notes after visiting some 20 plants of the British motor-vehicle industry: "At every hand we found evidence of the mechanization of work that had previously been done by manual methods."[50] Labor and machine costs in the United States showed the same trends between 1940 and 1950; both increased but labor costs increased at a much faster rate. Increases in relative labor costs (measured by the ratio of man-hour costs to kilowatt-hour costs) paralleled increases in mechanization

47 Seymour Melman, *Dynamic Factors in Industrial Productivity*, Oxford, England: Basil Blackwell, 1956, pp. 113–115.

48 *Ibid.*, p. 114. However, Figure 9 on this page is somewhat misleading as it combines data from different countries with data from the same country at different periods.

49 *Ibid.*, pp. 35–43, 97–99, 110–112.

50 *Ibid.*, p. 24.

(measured by horsepower installed per production worker) in the United States during the period 1899–1950. In short, rising costs of labor provide strong economic incentives for manufacturing firms to mechanize.

The higher the relative cost of labor, therefore, the greater is the productivity per man-hour.[51] For high labor costs promote mechanization, and mechanization improves productivity. This conclusion of Melman is supported by Abegglen's observation that the low labor costs in Japan divest the management of Japanese factories of incentives for seeking to increase productivity by introducing technological innovations.[52] Here we have an interesting paradox: if high labor costs encourage mechanization and enhance productivity, as they seem to do, high wages and, particularly, strong unions that can effect recurrent raises in wages and fringe benefits, although typically opposed by management, serve important latent functions for technological progress and hence for long-run productive efficiency.[53]

Shortages of materials may have the same latent function for improving production methods as high labor costs. This effect is illustrated by changes that occurred in a large manufacturing company studied by Richardson and Walker.[54] During World War II the plant was converted from job-lot to assembly-line production, a change which involved not so much the introduction of new machinery as innovations in the organization of the production process. A major factor in the decision to reorganize was the shortage of materials resulting from the war, which made job-lot production unsatisfactory, because it required that a large supply of parts and materials be kept on hand in store rooms, which workers could draw on when needed. Since the new assembly-line production method not only mitigated the specific problem of wartime shortages but also seemed to improve productivity, the shortages that stimulated the innovation may be considered to have had the latent function of promoting efficiency.

This century has witnessed considerable growth in the administrative apparatus of production organizations, both in absolute terms

[51] For various countries and time periods combined, Melman found a correlation between labor cost and productivity of +.88; *ibid.*, p. 125. (The data on the United States and Great Britain are presented on pp. 117–118.)

[52] Abegglen, *op. cit.*, pp. 110–120.

[53] In a later case study of an automobile manufacturing company, Melman shows how high labor costs constrained management to increase mechanization. Seymour Melman, *Decision-Making and Productivity,* Oxford, England: Basil Blackwell, 1958, pp. 105–106, 141–143, 153.

[54] Frederick L. Richardson, Jr., and Charles R. Walker, *Human Relations in an Expanding Company,* New Haven: Labor and Management Center, Yale University, 1948, pp. 54–74.

and in terms relative to the size of the production force.[55] Melman shows that increases in the ratio of the number of administrative personnel to the number of production workers paralleled increases in productivity per worker in both the United Kingdom and the United States over the period 1900 to 1950.[56] But notwithstanding this association, Melman argues that growth in the administrative apparatus did not contribute to productivity; on the contrary, it impeded it. He supports his thesis with three arguments.

First, he reports that the two trends are not closely parallel. Specifically, the comparison of twenty-one manufacturing industries in the United States reveals little relation between the increase in the number of administrative personnel and the increase in productivity during the first few decades of this century. But these are very rough and indirect measures, and the trends are undoubtedly affected by other factors. Note that Melman found that comparisons by industry revealed no correlation between mechanization and productivity, as we have seen, just as he found they showed no correlation between administrative apparatus and productivity. In the first case, however, he disregards the negative finding, explaining that it is due to the interference of other factors, and concludes that mechanization and productivity are related; whereas in the second case he accepts the negative finding and concludes that administration and productivity are unrelated. But he cannot have it both ways. Differences in type of organization between industries are just as likely to be responsible for the absence of a correlation between administrative expansion and increased productivity as for the absence of a correlation between mechanization and productivity. If firms within an industry were compared, one might well find an association between administrative services and productivity, just as one does for the trend in all manufacturing industries, because an administrative staff makes possible a more complex organization of work processes.

Second, Melman reports that large differences in productivity between the United States and the Soviet Union are not accompanied by similar differences in proportion of administrative personnel. This difference, however, merely shows that conditions other than administration influence productivity, a fact that is hardly open to question.

Finally, Melman cites evidence that output (specifically, value added to the product through manufacturing) per production worker has increased more rapidly between 1907 and 1948 than has output

55 See Seymour Melman, "The Rise of Administrative Overhead in the Manufacturing Industries of the United States, 1899–1947," *Oxford Economic Papers,* 3 (1951), pp. 64–66.

56 See Melman, *Dynamic Factors in Industrial Productivity, op. cit.,* pp. 132–140.

per employee (including administrative as well as production personnel). This finding, however, apparently reflects primarily the disproportionate increase of administrative personnel in recent years and does not do justice to the indirect contribution administration makes. The question remains: how much of the increased efficiency of today's production worker has been made possible by the more extensive administrative apparatus? This question is not easily answered, because the contribution made to production by administration is necessarily indirect.

The data Melman cites do not, in our opinion, warrant the conclusion he draws. Pending empirical research explicitly designed to deal with this problem, it remains a moot question whether or under which conditions an enlarged administrative machinery contributes to productive efficiency.[57]

Interorganizational Processes

Ecological Processs and Organizations. What are some of the social processes that characterize the relations between organizations? Ecologists have distinguished between two basic types of relations: symbiosis and commensalism. The former term denotes relations of mutual dependence and advantage between unlike structures; the latter refers to competition between like structures for the scarce resources needed for survival and growth.[58] In this section we present an analysis of competitive and exchange processes that occur between organizations, of the ways in which these processes change like structures into unlike ones and vice versa, and of the resulting dynamics in the larger social system. In a sense, this is an analysis of the ecology of organizational life.

Urban growth is frequently conceptualized as being governed by processes of competition for optimum location.[59] Since all organizations are dependent to some extent on their social environment, there is competition among them for the most desirable locations. Thus, a business firm is interested in a location close to a large labor market, to its suppliers, and to its customers, but it will find itself competing with other firms for these advantages. Ready access to exchange partners makes a central location in the city particularly

[57] For another discussion of the relation between technology and administration see James D. Thompson and Frederick L. Bates, "Technology, Organization, and Administration," *Administrative Science Quarterly,* 2 (1957), pp. 325–343.

[58] Amos H. Hawley, *Human Ecology,* New York: Ronald, 1950, pp. 36–39.

[59] See Robert E. Park *et al., The City,* Chicago: University of Chicago Press, 1925.

desirable for many types of firms. The outcome of competition for a central location depends on two factors: (1) the significance that being easily accessible to a wide public has for the firm, which is greater for a retail store than for a manufacturing concern, since only the former depends on being personally contacted by many customers; and (2) the firm's capacity to purchase real estate in high demand which commands a high price. These competitive processes lead to the dominance of certain types of organizations in various areas of the city. For example, large stores and offices typically dominate the downtown business district of the city, while manufacturing firms cluster around its outer boundaries.[60] The dominant firms exercise control over the area, not by ordering or forbidding but by determining the conditions under which other types of firms can survive and thrive.[61] The lesser firms must settle for second-best locations unless they can establish a symbiotic relation with the dominant ones. For example, there are luncheonettes on Wall Street but no factories; powerful financial firms welcome the presence of the former because it is to their advantage to have restaurants easily accessible to their large office staffs, while no advantage would accrue to them from the proximity of factories.

The conclusion of human as well as plant ecology is that competition leads to the dominance of one type, and once dominance has been established competition between types subsides. Classical economics, in contrast, assumes that competition helps to maintain and simultaneously is perpetuated by an equilibrium between many competitors in the market. What accounts for these opposite conclusions? The crucial difference between the two underlying conceptions of competition is that ecology takes as its unit of analysis the species, in our case a *type* of organization; whereas in classical economics the competing unit is not the species but the individual, that is, a specific firm. Although there are several important differences between the species and the individual, the distinction that is most relevant for an examination of the consequences of competition is the *potential expandability* of each unit. An example from plant ecology illustrates this point. The very nature of a tree sets inherent limits on its growth, but there are no such *internally* determined limits to the expansion

60 It is recognized that this generalization is broad and imprecise. There are many types of stores, offices, and manufacturing establishments and their varying needs create different location pressures. See, for example, Edgar M. Hoover and Raymond Vernon, *Anatomy of a Metropolis,* Cambridge, Mass.: Harvard University Press, 1959, pp. 25–124.

61 See the discussion of dominance in Donald J. Bogue, *The Structure of the Metropolitan Community,* Ann Arbor: Horace H. Rackham School of Graduate Studies, University of Michigan, 1949, pp. 10–13.

of a species of trees. Thus, disregarding external limitations, a whole continent could conceivably be covered with oaks as the dominant species; but a single oak tree could not possibly grow so large as to dominate an entire continent. This example suggests the general hypothesis that in the absence of internal limits to the expansion of a competitor, competition is likely to result in the dominance of one or a few related units. Whether the outcome of competition between organizations approximate the dominance model of the ecologist or the equilibrium model of classical economics, therefore, is expected to depend on the degree of expandability of organizations.

How expandable are formal organizations? Their expandability is probably not unlimited because some restrictions are imposed on it by requirements of their internal structure.[62] But these limits to organizational expandability appear to be very broad ones, as is indicated by the tremendous variations in size of firms. According to the hypothesis, therefore, the result of competition among formal organizations should be closer to dominance than the model of classical economics implies. The individual firm can expand to such a degree that one or a small number of firms can dominate the system in its area of specialization, that is, a particular market. This, indeed, is the conclusion of several more recent studies by economists who have rejected Marshall's dichotomy between pure competition and pure monopoly. Both Chamberlin and Robinson suggest that the typical end product of competition is a state of monopolistic competition or oligopoly—the domination of the market by a few strong firms.[63] The hypothesis advanced also helps to explain the increase in oligopoly-dominated markets during the last century. In earlier years, the size of the individual or family fortune set practical limits on the expandability of firms. But the institution of the share corporation has removed these financial limits to expansion.[64] Other innovations, such as improved communication methods and better managerial tech-

[62] Boulding suggests that the principle of nonproportional change governs the growth of organisms and organizations. This principle holds that a uniform increase in the linear dimensions of a structure will increase its area as the square, and its volume as the cube. Although some compensation for these changes is possible, Boulding holds that "the size of the structure itself is limited by its ultimate inability to compensate for the non-proportional changes." See Kenneth E. Boulding, "Toward a General Theory of Growth," *Canadian Journal of Economics and Political Science,* 19 (1953), pp. 326–340.

[63] Edward Chamberlin, *The Theory of Monopolistic Competition,* Cambridge, Mass.: Harvard University Press, 1933; and Joan Robinson, *The Economics of Imperfect Competition,* London: Macmillan, 1933.

[64] "Just as the factory system brought large numbers of *workers* under unified direction and control, so the corporate system brought numerous bits of *wealth* together under a single management." Robin M. Williams, Jr., *American Society* (2d ed.), New York: Knopf, 1960, p. 182 (italics in original).

niques, have further facilitated organizational expansion, as Boulding notes.[65] As the limits to the expandability of economic organizations have widened, the chances that one or a few would dominate a given market have increased.

Competition and Exchange. Competition, of course, is not confined to economic markets.[66] Political parties compete for votes, fraternal and religious organizations for members, participants in a discussion for speaking time, and group members for popularity and respect. Social competition, whether between organizations or persons, always involves some kind of exchange relations with other organizations or persons. For example, a firm competes with other firms for the patronage of still other firms or individuals. Success in social competition, therefore, depends on the ability of the firm to establish symbiotic relations in which extensive advantageous exchanges take place. Social competition should be distinguished from other forms of competition that do not involve complementary symbiotic relations, exemplified by plants competing for sunlight. Exchange relations are required for competition if its immediate object is not a natural resource but some kind of output or product of another unit in the system.

In rudimentary competitive social structures, the distinction between competitive and exchange relations does not yet exist. For example, the federal agents discussed in Chapter V competed *with* one another for the respect *of* one another. An agent's exchange partners, whose output of respect was the object of the competition, were identical with his competitors for superior informal status. However, once differentiation in respect developed, and only then, exchange relations became differentiated from competitive ones. Those whose competent advice won them the respect of their colleagues continued to compete with one another for superior status—for power and informal leadership. This competition furnished the inducement for them freely to give advice in exchange for deference. On the other hand, those whose advice did not earn them the respect of their colleagues had no chance in this continuing competition for high status: since their advice was not in demand, they could not exchange it for deference. These agents, then, became the potential "customers" of those whose earlier success enabled them to continue to compete for superior status. The low-status agents' output of respect and deference (for which they received help and advice) determined the outcome of the high-status agents' competition for top status. In sum, the

65 Kenneth E. Boulding, *The Organizational Revolution,* New York: Harper, 1953, pp. 25–32.

66 "Even the most sacred institutions are subject to the necessity of competing, in some measure, to survive." Everett C. Hughes, "The Ecological Aspect of Institutions," *American Sociological Review,* 1 (1936), p. 188.

outcome of earlier competitive processes leads to a differentiation in the social structure which has the result that some remain competitors and others become exchange partners in subsequent stages of competition.

These processes in which competitive and exchange relations become differentiated are also characteristics of stratification systems of entire communities. The class structure of the Ifugao tribes in the Philippines illustrates this.[67] There are three broad social classes among the Ifugao, based primarily on wealth. Everybody competes for wealth, and the relative success of an individual governs his social status. The middle class is composed largely of property holders who work on the land. The lower-middle class is composed of people whose land holdings are barely sufficient to provide a living and who are threatened by bankruptcy, which would throw them into the propertyless lower class. The upper-middle class, on the other hand, consists of families whose land provides a surplus income and who, therefore, have some chance of moving into the upper class. In effect, the members of the lower-middle class compete for staying in the middle class, while those of the upper-middle class compete for entry into the upper class. And exchange relations develop between these two strata within the middle class which help both partners in their respective competitive struggles. Members of the upper-middle class often make loans at interest to those of the lower-middle class, and this exchange process aids both in achieving the object of their competition. For the loan enables the poorer family to retain its land and thus its middle-class position, and the interest increases the wealth of the richer family and thus its chances of moving into the upper class. Greater advantage seems to accrue to the lender than the borrower in this exchange; this difference is not atypical for symbiotic relations that involve partners of unequal status and power. The reader may have noticed that the relations and exchange processes existing among the Ifugao are not unique but can be observed in some other societies as well. The point of major interest is that here again the outcome of earlier competition has differentiated between exchange relations and those that continue to be competitive.

Dominance involves an extreme form of this general tendency. If a few organizations become dominant in a market, symbiosis comes to replace competition as the central principle that governs relations between units in this larger system. This change is well illustrated by our earlier example taken from urban ecology concerning the dominance of the financial organizations in Wall Street. The entry of

[67] Irving Goldman, "The Ifugao of the Philippine Islands," Margaret Mead (ed.), *Cooperation and Competition among Primitive Peoples,* New York: McGraw-Hill, 1937, pp. 153–179.

other types of organizations into this area is no longer governed by competitive processes but by a need for their services. Lunch counters are found in this area not because they were able successfully to compete with the financial companies but because of their symbiotic relation with this dominant type of organization in the area. The same principle applies to economic markets. In an industry dominated by a few major companies, such as the automobile industry, the success of the smaller firms depends not on successfully competing with these giants but on establishing symbiotic relations with them, for example, as suppliers of various kinds of parts. It follows that the keystone to understanding these new markets is no longer an analysis of competitive processes but an analysis of the interdependence and power relations between the units in the system. This conclusion is similar to one Veblen reached long since,[68] and it is the focus of Galbraith's theory of countervailing powers.

Galbraith's main thesis is that the primary regulating mechanism of the modern market is no longer competition but rather the structure of power relations between exchange partners. In his own words:

> The long trend toward concentration of industrial enterprise in the hands of a relatively few firms has brought into existence not only strong sellers . . . but also strong buyers . . . The fact that a seller enjoys a measure of monopoly power, and is reaping a measure of monopoly return as a result, means that there is an inducement to those firms from whom he buys or those to whom he sells to develop the power with which they can defend themselves against exploitation . . . Competition which, at least since the time of Adam Smith, has been viewed as the autonomous regulator of economic activity and as the only available regulatory mechanism apart from the state, has, in fact, been superseded . . . In the typical modern market of few sellers, the active restraint is not provided by competitors but from the other side of the market by strong buyers.[69]

The processes described by Galbraith are well illustrated in a government survey of United States business.[70] This report shows how the dominant market position of wholesalers was undermined at the end of the last century by the increasing power of manufacturing concerns employing mass-production methods. At the same time retailers, with the establishment of mail-order houses and chain stores, also became more powerful. The wholesalers, squeezed from both sides,

68 Thorstein Veblen, *The Theory of Business Enterprise,* New York: Scribner, 1904.

69 John Kenneth Galbraith, *American Capitalism,* Cambridge, Mass.: Houghton, Mifflin, 1952, pp. 118–119.

70 Temporary National Economic Committee, *Investigation of Concentration of Economic Power: The Problems of Small Business,* Washington, D.C.: U.S. Government Printing Office, 1941 (Monograph No. 17).

consequently found themselves seriously threatened. In order to increase their power and meet these threats, they established trade associations and lobbied successfully for legislation that penalized chain stores. These actions of wholesalers elicited counter-actions by retailers and manufacturers. The former established trade associations of their own and developed larger outlets, such as supermarkets, to escape the tax on chain stores. Manufacturers also united by forming trade associations and organized sales departments to bypass wholesalers. Simultaneously, manufacturers and retailers were engaged in their own power struggle, the former relying on product differentiation, branding, and advertising to force retailers to carry their products, the latter reacting by using their own brand names or even manufacturing their own brands to become more independent of manufacturers. In sum, as a few large organizations became dominant in each market, the focus of their endeavors shifted from competing with others in the same market to struggling over the conditions and terms of exchange between buyers and sellers, a struggle that induced former competitors to become allies and organize mutual-benefit associations.

Looking at these processes of development from another perspective, one can see that the outcome of competition at one stage changes the object of the competition at the next. In the earlier stages, all units compete against all others. Differential success in this competition for relative standing has the result that the least successful are no longer able to compete with the most successful, and this change leads to the establishment of symbiotic relations that aid both parties in their continued competition. But now the two groups have different objects in their competition: the more successful units compete with one another for dominance, while the less successful units compete with one another for survival, as illustrated by the Ifugao middle class as well as in economic markets. Relations between units have also changed, since some former competitors have now become partners in exchange relations. Differential success in further competitive processes enables a few large organizations (or other units) to dominate a given market. At this advanced stage when dominance has been achieved, a new object of competition emerges, namely, the ability to exercise controlling power in exchange relations. The consequent power struggle between buyers and sellers puts the members of each group under pressure to unite by establishing agreements and trade associations. In this manner former competitors have once again been transformed into partners. But the reverse change has also occurred since organizations that were formerly partners in exchange relations—the buyers and sellers—have become competitors for dominant power over a set of interrelated markets. This latter competition for power in exchange relations that link markets has, according to Galbraith, re-

placed the earlier competition for profit or dominance in one market as the main regulatory system of the American economy.[71]

The processes in which competitive relations become exchange relations and exchange relations turn into competitive ones furnish a graphic illustration of the dynamics that characterize the social context of organizational life.

Concluding Remarks

In preceding chapters of this work we have been concerned largely with the internal structure of formal organizations. We have analyzed the structure of colleague relations and of hierarchical ones, and we have discussed the relations between members of an organization and its public. But in this chapter our focus has shifted to the social context within which organizations function. We have briefly examined organizations in differing communities and cultures in order to note some of the interrelations between organizations and their environments. However, the social context is not merely an aspect of the external environment but actually penetrates organizational life. For example, the character of the labor force or the degree of technological complexity can be treated either as an attribute of the environment or as a characteristic of the organization itself. We examined some comparative studies to infer the nature of the interdependence between various basic characteristics of organizations. Finally, we turned to the relations between organizations and analyzed the processes that govern these relations and the dynamic changes in the larger social system resulting from these processes.

The internal dynamics of organizational life has been implicit in much of our earlier discussion. In our final chapter we shall make it our explicit focus as we summarize some of the main points of the previous analysis.

71 It is, of course, possible that the outcome of the competition between countervailing powers is that one supertrust comes to dominate the entire economy. In this case, the only remaining object of competition would be world domination.

CHAPTER NINE

Organizational Dynamics

This concluding chapter focuses attention on the internal dynamics of organizational life. We shall summarize the main points of our foregoing analysis with particular emphasis on the processes of change in formal organizations, processes that have sometimes been left implicit in the earlier discussion. To do justice to problems of organizational change, however, we shall not confine the analysis to material already presented in previous chapters but shall draw upon additional relevant sources whenever appropriate.

The concept of dilemma greatly contributes to an understanding of internally generated processes of change in a social organization. If we conceive of social systems as meeting basic needs and requisites, the implicit assumption is that adjustment prevails and change does not occur unless new *external* conditions require adjustments. But if we conceive of social systems as being confronted by dilemmas, that is, by choices between alternatives in which any choice must sacrifice some valued objective in the interest of another, the implicit assumption is that problems are endemic and, therefore, serve as a continual *internal* source of change in the system. The concept of dilemma by no means implies that improvements are not possible, but it does imply that final solutions and perfect adjustments are impossible. Dilemmas confront not only formal organizations, as we shall see in this chapter, but also the *study* of formal organization, as was mentioned in the first chapter.

A fundamental methodological dilemma in the study of social organization is posed by the necessity to investigate the interdependence between units in a larger social system, on the one hand, and the requirement to treat subunits of the totality under investigation as independent cases in order to derive generalizations about them, on the other. The investigator must choose between alternative approaches and invariably sacrifice one of these requirements. Research that is being conducted in one formal organization, for example, may either treat the various sections in the organization as independent units for analysis or focus upon their interdependence in the larger organization. The first alternative furnishes independent cases that permit deriving generalizations about work-group organization, but it ignores the

principles that govern the ways in which work groups are organized into a larger formal structure. The second approach deals with these interrelations between sections that characterize the formal organization, but it thereby sacrifices the independent cases required for generalizing. The comparative study of many organizations would solve this specific problem—that is, make it possible to advance and test generalizations about formal organizations—but the generic dilemma would reappear as one raises the question of what principles govern the interrelations between these organizations in the larger society.

The comparative approach in the study of formal organizations offers probably the best heuristic solution for this methodological dilemma at this stage of knowledge in the social sciences. Ideally, this approach would involve a research design in which organizations in a sample are systematically compared. Since such studies are not as yet available, however, we have had to be content with approximating this ideal as best we could with less systematic comparisons. We have relied primarily on three methods: secondary analysis comparing case studies of various organizations; internal comparisons of different segments in one organization; and comparison of two organizations with many similarities—the two welfare agencies. We have tried to use the comparative approach in this book extensively, even though the cases we have examined were not always clearly comparable and many conditions had not been controlled. Hence, all of our conclusions are inferential and must be considered only suggestions, not confirmed propositions.

These problems are even more acute when we turn to the analysis of the processes of change in organizations. Systematic studies of organizational change are virtually nonexistent; therefore, the conclusions we reach about the dynamics of organizational life are mere hypotheses awaiting validation in future systematic research.

Organizational Development

Development of Organizational Complexity. Perhaps the minimum definition of formal organization is that collective effort is explicitly organized for specific ends. We have distinguished such formally established organizations from crescive organizations which emerge in the course of social interaction without explicit intent or deliberate design. Formal organizations vary in size and complexity from a modern army at one extreme to the collectively organized recurrent fishing expeditions found in nonliterate tribes at the other. Since the latter involve formally established procedures to accomplish an explicit objective, albeit rudimentary ones, they are included in our

concept of formal organization. To be sure, the larger and more complex formal organizations are of special interest to students of contemporary social organization, partly because they are so prevalent today, partly because special problems are often posed by the bureaucratic apparatus that tends to develop with increasing organizational size and complexity. The typical characteristics of such bureaucratic machinery, according to Weber, include the division of work into areas of specialized jurisdiction and competence; a hierarchical authority structure; general rules and regulations that govern operations; an emphasis on impersonal detachment designed to eliminate personal considerations from official decisions; and personnel policies based on objective criteria which encourage commitment to a full-time lifelong career.

Formal organizations can be classified in many different ways. We have proposed a classification based on the criterion *cui bono*—who the prime beneficiary is in whose interest the organization is expected to operate. The resulting four types of formal organizations are: (1) mutual-benefit associations, such as unions, which are intended to promote the interest of rank and file members; (2) business concerns, such as factories, which are established to serve the interest of their owners; (3) service organizations, such as social-work agencies, which are expected to be governed by the welfare and interest of their clients; and (4) commonweal organizations, such as police departments, which function to further the welfare of the public-at-large. If another group usurps the role of the prime beneficiary in any of these types, the objectives of the organization are thwarted and its original function is distorted or changed.

The fact that an organization has been formally established does not mean that all the social processes within it follow predesigned patterns. Large and complex formal organizations do not spring into existence full-blown but develop out of simpler ones. We shall examine such developmental processes in this first part of the chapter. Moreover, the formally institutionalized mechanisms for coordinating collective efforts merely set the limits within which social processes become informally organized. The next part of the chapter is devoted to these informally emerging patterns in formal organizations and their significance for official operations.

Some inferences can be drawn concerning the early development of organizational complexity from Udy's study of production organizations in nonliterate societies.[1] His cross-cultural comparisons suggest

[1] See, in particular, Stanley H. Udy, Jr., "'Bureaucratic' Elements in Organizations," *American Sociological Review*, 23 (1958), pp. 415–418, and "Technology, Society, and Production Organization," paper read at the meetings of the American Sociological Association, New York, 1960.

that a hierarchy of authority, in which persons in the lower levels are dependent for their rewards on those in the higher levels, tends to develop early in simple production organizations. There is also some indication that such rudimentary bureaucratic hierarchies are rooted in the ascribed status system of the community, inasmuch as these hierarchies were more frequently found in organizations where membership was based on kinship or political status rather than on a voluntary contract. Specialization typically evolves only in organizations already characterized by hierarchical differentiation, and it, in turn, tends to precede the development of more complex institutional arrangements, such as gearing rewards to contributions, and introducing contractual agreements to replace community status as the basis for defining the terms of membership participation. It would appear, then, that an organizational hierarchy rooted in an ascribed status system furnishes the initial impetus for the development of organizational complexity, but further developments tend to make participation in the organization independent of the community's ascribed stratification system. As organizations become more complex and differentiated in internal structure they also seem to become more differentiated from the larger societal structure, forming relatively autonomous subsystems with their own principles of organization.

Bureaucratization. Structural growth by its very nature involves increasing complexity. Boulding derives this conclusion from his "principle of non-proportional change": since the rates of growth of the various parts of an organization are not proportional, growth always entails internal adjustment and change.[2] One of the most important changes that occurs as organizations become larger and more complex is the development of an administrative apparatus. Many observers have lamented the trend toward larger administrative overhead in organizations as indicative of overbureaucratization. Parkinson has satirized the presumably parasitic character of administrative personnel most wittily, suggesting that the less work there is in an organization, the greater are the increases in its administrative staff. He cites in support of his "law" the fact that while the number of officers and men in the Royal Navy decreased by 31 per cent between the years 1914 and 1928, the number of Admiralty officials increased by 78 per cent.[3] Haire also comments on "the remarkable resistance of the staff to negative growth," noting that in 19 cases of layoffs observed in four different firms only line workers were involved and that in a few

2 Kenneth E. Boulding, "Towards a General Theory of Growth," *Canadian Journal of Economics and Political Science,* 19 (1953), pp. 326–340.

3 C. Northcote Parkinson, *Parkinson's Law and Other Studies in Administration,* Boston: Houghton-Mifflin, 1957, pp. 7–8.

instances new staff personnel was hired during the layoff period.[4] Powerful administrators undoubtedly can and often do protect their positions against being affected by reductions in staff. Parkinson's illustration, however, is quite misleading, since it unquestionably reflects in part technological advances in warfare and the changes that occur in a military service as it adjusts to peacetime—conscripts are discharged and regulars are gradually promoted, since it is important to maintain a core of staff officers for future emergencies.

It is widely assumed that large organizations tend to be overbureaucratized, that is, that an increase in organizational size is accompanied by a disproportionate increase in administrative overhead; but the evidence does not support this assumption. To be sure, the average size of manufacturing firms has increased during the first half of this century, and so has the proportion of personnel devoted to administration rather than production.[5] These trends, which reflect basic changes in the organization of industrial concerns, may be responsible for the prevailing impression that size is associated with overbureaucratization. The growth of government services during the last few decades may also have contributed to this notion. Except for small firms, however, increases in size are not associated with increases in the proportion of administrative personnel. Only during an organization's early stages of growth does the proportion of administrative officials increase;[6] further growth is not accompanied by increases in administrative overhead. To cite specific cases, a study of 211 manufacturing firms in Ohio conducted by Baker and Davis found no relation between size of organization and proportion of administrative officials.[7] Melman's data on American manufacturing concerns even reveal an inverse relation between size and proportion of administrative personnel.[8] Data on German industries analyzed by Bendix also show an inverse relation between size and proportion of administrative staff; in 1933, for instance, the proportion of administrators was lower in firms employing more than 1,000 employees than in those employing between 51 and 200 workers.[9]

[4] Mason Haire, "Biological Models and Empirical Histories of the Growth of Organizations," Mason Haire (ed.), *Modern Organization Theory,* New York: Wiley, 1959, pp. 292–293.

[5] Seymour Melman, "The Rise of Administrative Overhead in the Manufacturing Industries of the United States, 1899–1947," *Oxford Economic Papers,* 3 (1951), pp. 64–66, 89.

[6] Haire, *op. cit.,* pp. 288–292, 305.

[7] Alton W. Baker and Ralph C. Davis, *Ratios of Staff to Line Employees and Stages of Differentiation of Staff Functions,* Columbus: Bureau of Business Research, Ohio State University, 1954, pp. 14–15.

[8] Melman, *op. cit.,* pp. 89–90.

[9] Reinhard Bendix, *Work and Authority in Industry,* New York: Wiley, pp. 221–222. Bendix shows that the proportion of technicians does increase as a function of size of firm; but since administrative officers are the larger group, "the over-all tendency remains for bureaucratization to be highest in the smaller firms."

In industrial organizations, then, bureaucratization as indicated by proportion of administrative personnel is, contrary to prevailing impressions, not directly related to size, and may even be inversely related to it. Two studies of other types of organizations have also dealt with this problem. Terrien and Mills present the only systematic data confirming the popular conception: the size of school districts in California was directly related to the proportion of administrative personnel, although the relations observed were small.[10] Anderson and Warkov, on the other hand, examining 49 Veterans' Administration hospitals, found an inverse relation between the size of a hospital and the proportional size of its administrative staff.[11] Their analysis also suggests that complexity[12] as distinguished from size, although the two usually go together, is directly associated with the proportion of administrative personnel, and this distinction provides a possible reason for the discrepancy between the results of the school-district study and those of other research on this problem. Larger school districts were probably more complex than smaller ones—administering several schools in different locations rather than a single one—and this complexity, not size itself, may have been responsible for their larger administrative staffs. The conclusions that emerge from the research findings are that large organizations do not typically have disproportionately large administrative machineries; that, however, size tends to be directly related to complexity, and complexity to a large proportion of administrative personnel; and that the size of an organization, particularly if complexity is held constant, may actually be inversely related to the relative size of its administrative staff.

In mutual-benefit associations bureaucratization poses the special problem of oligarchy. The functions of this type of organization, where the members are expected to be the prime beneficiaries and to govern themselves, are placed in jeopardy by the development of a bureaucratic apparatus that centralizes power in the hands of administrative officials, as was shown in Michels' famous study of unions and social-democratic parties in Germany.[13] The egalitarian ideology and objectives of these organizations would lead one to expect them to be democratically governed by their membership. But the need for efficient administration to insure success in bargaining or in elections

10 Frederic W. Terrien and Donald L. Mills, "The Effects of Changing Size upon the Internal Structure of Organizations," *American Sociological Review,* 20 (1955), pp. 11–13.

11 Theodore R. Anderson and Seymour Warkov, "Organizational Size and Functional Complexity," *American Sociological Review,* 26 (1961), pp. 23–28.

12 Tuberculosis hospitals were considered noncomplex, whereas general hospitals, which serve other kinds of patients as well as tubercular ones, were considered complex. See *ibid.,* pp. 25–27.

13 Robert Michels, *Political Parties,* Eden Paul and Cedar Paul (trans.), Glencoe, Ill.: Free Press, 1949, (first published 1915).

encourages the development of a bureaucratic machinery with effective control of the organization centralized in the leadership. Moreover, as the experience gained by the leaders makes them virtually indispensable for the successful implementation of organizational objectives, their dominant position is further strengthened. The examination of such consequences led Michels to propose his "iron law of oligarchy," according to which even initially egalitarian organizations invariably develop hierarchical structures in the interest of effective accomplishment of objectives. There can be little doubt that Michels directs attention to widely prevailing tendencies in mutual-benefit associations. The same developments that he describes have characterized the history of many American unions. Contrary to his assumptions, however, these developments, though prevalent, are not inevitable, as the study of the International Typographical Union by Lipset and his colleagues demonstrates.[14] This union has successfully resisted domination by a self-perpetuating oligarchy, because it has maintained a two-party system which safeguards the democratic election of union leaders.

Bureaucratization entails the danger that the original objectives of the organization are lost sight of as the result of preoccupation with administrative problems. Thus, Michels observes that the radical programs of socialist unions and parties became increasingly modified and conservative once bureaucratic hierarchies had developed. For the leaders, interested in preserving and increasing the organization's strength, willingly abandoned radical objectives in favor of more moderate ones that did not threaten the organization's survival in a hostile society. Note the dilemma implied here: a party or a union must build a strong organization and assure its survival to achieve its objectives, yet preoccupation with such organizational problems leads to the surrender of these very objectives. Modifications of this kind in the organization and its program occur not only in mutual-benefit associations but also in other types of formal organizations.

Selznick's analysis of the process of cooptation illustrates how a different mechanism brought about similar changes in a government agency, namely the Tennessee Valley Authority.[15] The reform program of the TVA encountered strong opposition from powerful entrenched forces in the area, and the TVA's grass-roots policy required that it achieve success by coming to terms with these opposition forces rather than impose its will upon them by relying on the power of the federal government. As a means for adjusting to its hostile environment, the TVA coopted some representatives of the opposition into its manage-

[14] Seymour M. Lipset *et al., Union Democracy,* Glencoe, Ill.: Free Press, 1956, pp. 201–269.

[15] Philip Selznick, *TVA and the Grass Roots,* Berkeley: University of California Press, 1949.

ment. However, since these new elements helped now to shape the policies of the organization in accordance with their own interests, the earlier objectives of the TVA were modified and transformed. For example, an initial policy of the TVA was to purchase large land strips around the reservoirs created by its dam-building operations in order to enable the public to benefit from the increased value of this land resulting from the expenditure of public funds. But the representatives of local agricultural interests who had been coopted into the policymaking bodies of the TVA succeeded in reversing this policy and adopting the procedure that only the minimum amount of land required for reservoirs and other facilities be acquired, thus enabling private investors to reap the benefits from the improved surrounding land.[16] Again, changes introduced to promote the adjustment of an organization to a hostile environment modified the organization's objectives.

The general principle illustrated by these changes is that in the course of adopting means to attain organizational goals the means may become ends-in-themselves that displace the original goals. In an innovating organization, whether political party, union, or government agency, this displacement tends to take the form, as we have just seen, of a retreat from the initial program to a more moderate and conservative program in the interest of maintaining the strength of the organization in an adverse environment. An example of this process in a voluntary association is furnished by Messinger, who describes how the fund-raising activities of the Townsend Organization were transformed from means into ends-in-themselves as new social conditions rendered its original objectives obsolete.[17] But in old and established bureaucracies the displacement of goals typically assumes a different form—the one to which Merton referred when he introduced the concept—namely, a rigid conformity with official procedures at the expense of the objectives they are designed to accomplish, which results from bureaucratic pressures and an overemphasis on discipline. In short, displacement of goals underlies the very tendencies associated with the stereotype "bureaucracy." In Merton's own words:

> Adherence to the rules, originally conceived as a means, becomes transformed into an end-in-itself; there occurs the familiar process of *displacement of goals* whereby "an instrumental value becomes a terminal value." Discipline, readily interpreted as conformance with regulations, whatever the situation, is seen not as a measure designed for specific purposes but becomes an immediate value in the life-organization of the bureaucrat. This emphasis, re-

[16] *Ibid.*, pp. 196–204.

[17] Sheldon L. Messinger, "Organizational Transformation," *American Sociological Review,* 20 (1955), pp. 3–10.

sulting from the displacement of the original goals, develops into rigidities and an inability to adjust readily.[18]

The very opposite processes, however, have also been observed in bureaucracies. For example, the orientations of officials in a government agency established to enforce New Deal legislation revealed a decade later not an inclination to retreat from the original reform program but a tendency to use it as a steppingstone for further reforms.[19] The earlier goals were not displaced by means that became ends-in-themselves but rather were succeeded by more advanced objectives. Officials advocated legislation that would make the agency responsible for administering more advanced reforms. While officials were in part guided by an idealistic attachment to furthering the program of the New Deal, their career interests provided additional strong incentives for their support of new programs. For once the original mission of getting the new laws generally accepted was accomplished, the task of continuing enforcement involved narrower and more routine responsibilities. These tasks were not as interesting as instituting new reforms and, furthermore, could be accomplished with a smaller staff. Indeed, reductions of staff had already taken place and more threatened to follow. New reform programs would create new challenges that would make the work more interesting and necessitate staff expansions that would avert the threat of layoff and improve the promotion chances of the present personnel. The economic as well as the psychological interests of these officials motivated them to advocate new reform objectives.

This succession of goals, the opposite of the displacement of goals, has also been observed in labor unions. Hart's study of a local of the United Automobile Workers in Windsor shows that this union had expanded its activities far beyond the function of representing workers in relation to management by providing a variety of services to members, such as loans, counseling, consumer cooperatives, and health care.[20] This proliferation of goals apparently occurred when the accomplishment of the union's original objectives had become routine, because the industry was thoroughly organized and industry-wide collective bargaining was carried out by the international union. Local union leaders, consequently, had reason to fear that members might look upon them as dispensable, and this possibility threatened their position (just as the indispensability of union leaders strengthens

[18] Robert K. Merton, *Social Theory and Social Structure* (2d ed.), Glencoe, Ill.: Free Press, 1957, p. 199 (italics in original).

[19] See Peter M. Blau, *The Dynamics of Bureaucracy,* Chicago: University of Chicago Press, 1955, pp. 193–200.

[20] C. W. M. Hart, "Industrial Relations Research and Social Theory," *Canadian Journal of Economics and Political Science,* 15 (1949), pp. 53–73.